UNBLIND

Build plan for Callum

Dosing (acute) module: the agents that live inside the trial

Your mission tonight: Catch blinding failure while the trial can still fix it.

Why this is yours: Your physics and computer science background suits the module that is most like running an instrument: noisy measurements arriving daily, a detector that must fire on real drift and stay quiet otherwise, and evidence a scientist can audit. You also own the strongest live demo moment.

At a glance	
Deadline	Working prototype on main by 00:00 tonight. Write-up, pitch and rehearsal tomorrow.
Your teammates	Danny and Izzy build the other two modules in parallel. You only meet through the shared contract.
How to read this	Sections 1 to 5 are the same in all three plans: the whole project. Your personal sections follow, then the shared timeline, demo script and glossary.
Golden rules	Contract frozen at 20:00. Own your folder. Every agent works with no API key. Every decision is logged. Every trial-changing action goes to a human. Synthetic data is labelled.

1. The project in one page

UNBLIND is an agentic platform that runs a psychedelic clinical trial with a team of AI agents, from protocol design to the FDA report, so the trial measures and controls the bias that sinks these drugs today.

The problem in one breath

To prove a drug works, you compare a drug group with a placebo group and nobody knows who got what. Psychedelics break this: patients know they tripped, and so do the clinicians scoring them. Every result becomes drug effect plus expectation, stuck together. The FDA's final psychedelic guidance (July 2026) names this functional unblinding as the central threat to validity, and it was central to the 2024 rejection of the MDMA application.

Why it stays unsolved

Nobody owns blinding as a job. Most trials never measure it, the ones that do measure it differently, and problems are only discovered at the end, after the money is spent. Trials are also slow to fill, and screening volunteers is manual.

What we are building

A master agent coordinates three modules, one per phase of a trial. Each module is a small team of agents that does real work every simulated day and brings decisions to a named human.

Phase	What the agents do	Owner
Pre-dosing	Co-design the protocol (risk, fixes, drafted blinding section, re-check on every edit), write ethics-ready recruitment ads, pre-screen applicants automatically, build the standard measurement plan and schedule	Danny
Dosing (acute)	Run post-session check-ins (guess, confidence, expectancy), structure monitor debriefs, monitor blinding per site and per rater through a sealed vault, detect drift, propose actions, chase missing data, scan new literature weekly	Callum
Post-dosing	Standard follow-up collection, a researcher question portal with a bias gate, a harmonised data export, and the FDA blinding report with an expectancy-adjusted effect	Izzy
Master agent	Runs the clock, routes messages between agents, keeps the trial memory, queues human decisions, shows mission control	Danny

What the judges will see

- A judge pastes or edits a draft protocol. The pre-dosing agents score its blinding risk, fix it, and screen 260 synthetic applicants down to a clinician-approved shortlist.
- Press play. Simulated weeks pass in about two minutes: sessions, check-ins, debriefs and a live blinding chart per site.
- Halfway through, raters at one site start guessing right far too often. The dosing agents spot it, explain it and propose moving those raters to remote assessment. The judge clicks approve.
- The judge can play a patient and answer a check-in.
- At close-out the post-dosing agents produce a standard FDA blinding report: every measurement, alert and human decision, plus the reported and expectancy-adjusted effect.

Everything simulated is SYNTHETIC and labelled as such, which the hackathon rules allow. No real patient data, ever.

2. The bottlenecks, quantified and sense-checked

The brief says every claim must name a bottleneck, quantify it, and show exactly how we move it. These are the numbers we can stand behind tonight. Anything not listed here needs a source before it goes in the pitch.

Bottleneck	Number (with source)	How UNBLIND moves it	Metric we report
Blinding is rarely measured	Only 29.5% (33 of 112) randomised psychedelic trials assessed blinding (systematic review, JAMA Psychiatry 2026)	Every participant and rater gets the same scheduled guess check-in by default	Share of scheduled blinding check-ins completed
When measured, blinding fails	Psilocybin, LSD and ayahuasca studies frequently reported failure above 90%; inert-placebo MDMA trials above 85% (same review)	Risk predicted before the trial; drift caught per site and per rater within days	Days from drift onset to alert
Regulators now demand it	FDA final guidance, July 2026: blinding questionnaires, expectancy measures before randomisation and at end of treatment, central blinded raters, alternative controls	Protocol checked against each item; the evidence collected into one report	Guidance items satisfied, before and after
Sessions are labour-heavy	Two monitors per session and a physician within 15 minutes (FDA guidance)	Debriefs spoken or typed, structured automatically into adverse events and unblinding cues	Debriefs structured automatically
Effects are small	COMP005: -3.6 MADRS vs placebo (95% CI -5.7 to -1.5). COMP006: -3.8 vs 1 mg psilocybin (95% CI -5.8 to -1.8) (Compass press releases)	Close-out report shows reported and expectancy-adjusted effect side by side	Reported vs adjusted effect, with interval
Blinding data is not comparable	Of 18 real trials our research agent examined, only 4 report clean per-arm guess data; timing ranged from dosing day to day 365 (our candidate gold set, unverified)	One standard instrument set, UNBLIND-BLIND-1, at fixed timepoints, exported in one schema	Trials exporting in the standard schema
Recruitment and screening are slow and manual	NEEDS A SOURCE. Do not quote a number until Callum or Izzy finds one	Agent screens every applicant in seconds; clinicians review only a shortlist and the flagged cases	Share screened automatically; clinician reviews per enrolled participant; staff hours saved (state the per-applicant minutes as an assumption)

Sense checks: what a scientist judge will push on

- **Bigger trials do not reduce bias.** More patients shrink random error, not the systematic error from knowing your group. A 1,000-person trial with broken blinding is just precisely wrong. Pitch recruitment as faster and fairer, not as less biased. Also, phase 3 psychedelic trials are already large (COMP006 dosed 581), so say 'most academic phase 2 trials are small'.
- **Pre-screening cannot remove humans entirely.** A clinician legally confirms final eligibility, especially for psychosis history, heart conditions and medication. The agent automates the funnel so humans review only a shortlist. That is the honest efficiency claim.
- **Prior psychedelic use: balance it, do not exclude it.** Experienced users recognise the drug faster, but excluding them shrinks the pool. Stratified randomisation spreads them evenly across arms.

- **Targeted ads using health or drug-use interests are risky** under UK GDPR (special category data) and are restricted by ad platforms. We park targeting and build compliant, ethics-ready ads instead.
- **Researcher questions can themselves break blinding.** Asking about 'mystical experiences' before the primary result is locked tells patients what they got. Every question passes a bias gate and is released only after the primary lock.
- **Reverse causation.** Patients who improve may guess 'drug' because they improved, so correcting for guesses can remove real effect. The adjusted effect is a conservative bound, not the truth. Our synthetic test shows the size of this failure (about 0.15 g over-correction).
- **Facts to say correctly:** the 3.8 is COMP006 against 1 mg psilocybin, not placebo. 'Above 90% failure' applies to many psilocybin, LSD and ayahuasca studies, not to all trials. Compass has not been 'cleared': rolling review is a submission route, not a verdict.
- **Prior art:** Szigeti and colleagues published an 'equal unblinding' meta-analysis in 2026. Our novelty is the agentic, living, design-time platform, not the idea that blinding matters.

3. How the platform works

The shared foundations (the only things modules share)

Piece	What it is	Why it matters
TrialRecord	One JSON document per trial holding the protocol, applicants, schedule, check-ins, metrics, alerts, approvals and events. Each module writes only its own sections.	Lets three people build in parallel without touching each other's code
MessageBus	Agents post messages (for example 'rater_drift'). The master agent routes them to whoever needs to know.	Agents cooperate without importing each other
ApprovalQueue	Any action that changes the trial becomes a request. A named human approves or rejects. Everything is logged.	The brief's 'human signs off key decisions', made concrete
Vault	The only holder of who got the drug. Agents receive group-level statistics only, with groups under 5 suppressed. Unsealing needs a named person and is logged.	Security: the AI cannot leak what it never sees
World	A SYNTHETIC simulator of applicants, patients, sites and raters, with planted events (rater drift at Site B from day 28, missing data at Site C, exclusions hidden in free text).	Lets us demo months in minutes, and evaluate agents against a known truth

The master agent

- Advances the simulated clock one day at a time and sets the phase: design, dosing, follow-up, close-out.
- Delivers messages and decides who hears about what. For example, rater drift goes to pre-dosing (draft a protocol amendment) and to post-dosing (record it in the FDA report). With a Gemini key it reads the message and chooses recipients; without one it uses a routing table. Either way the choice is logged.
- Isolates failures: if one module crashes, the others keep running and the error appears in the timeline.

- Writes a weekly summary and powers mission control: phase bar, timeline, approval inbox, agent message map, KPIs.

The simulated timeline

Day	Phase	What happens	Module
Setup	Design and recruitment	Protocol parsed and reviewed, fixes proposed, ads written, 260 applicants screened, clinician approves shortlist, stratified randomisation in the vault, measurement plan and schedule built	Pre-dosing
1 to 20	Dosing	Staggered dosing sessions; monitor debrief the same day; participant check-in the next day	Dosing
Each participant's weeks 1, 3, 6	Dosing	Rater visit: MADRS score plus rater guess; participant check-in	Dosing
Daily	Dosing	Blinding metrics per site and rater, drift detection, missing-data chasing	Dosing
Weekly	All	Literature scan with human approval; master weekly summary	Dosing and master
After last week-6 visit	Primary lock	Dosing tells the master the primary endpoint is locked	Dosing to master
Weeks 8 and 12	Follow-up	Standard follow-up collection; researcher questions released through the bias gate	Post-dosing
Day 90	Close-out	FDA blinding report and harmonised export, signed off by a human	Post-dosing

What already exists and gets reused

- **The UNBLIND research pipeline** (already in the repo): eight agents that read any trial and extract its facts with verified quotes, a deterministic quote auditor, the meta-regression, the design advisor, a Streamlit app, an evaluation suite and 18 candidate real trials. In the platform it becomes the weekly literature agent (dosing) and the risk engine (pre-dosing and post-dosing).
- **A starter scaffold** drafted by Claude in its own workspace: core.py (record, bus, approvals, vault), world.py (simulator), master/orchestrator.py and MODULE_CONTRACT.md. It is not in the repo yet. Danny decides at kick-off whether to pull it in; Claude can sync it on request.

4. How this scores against the judging criteria

Criterion	What a top score needs	Where UNBLIND delivers it	Owner
Uniqueness	Beyond a literature chatbot or stock model	Agents that live inside a trial for months; a vault the AI cannot read; a bias gate on researcher questions	All
Real-world need	One specific bottleneck, named user, plausible next step	Unmeasured, failing blinding. User: the clinical development lead running a psychedelic trial. Next step: pilot the standard kit with one contract research organisation	Izzy (narrative)
Science, data and evaluation	Tested against a baseline, data referenced and licence-checked, uncertainty stated	Drift detection vs end-of-trial-only checking; screening vs synthetic ground truth; adjusted vs naive effect on planted-truth simulations; sources and licences listed	Callum and Izzy

Criterion	What a top score needs	Where UNBLIND delivers it	Owner
Agent design and live demo	Real multi-step agent; demo handles a judge's new input	Judge edits a protocol, approves actions, plays a patient, submits a researcher question	Danny (shell), everyone (moments)
Security and responsible design	Named risks with mitigations; human signs off key decisions	Vault, approval queue, quote audit, bias gate, consent checks, GDPR notes, synthetic labelling	All

5. Who does what

Person	Owns	Mission
Danny	Pre-dosing module, master agent and mission control (platform lead)	Make the trial start right, and make the whole platform run as one product.
Callum	Dosing (acute) module: the agents that live inside the trial	Catch blinding failure while the trial can still fix it.
Izzy	Post-dosing module, the FDA blinding report and the science of the evaluation	Turn everything the trial collected into evidence a regulator and a scientist can trust, and make that evidence comparable across trials.

Callum, the next sections are yours. Everything below is written for your module. Your teammates have the same shared sections and their own versions of these.

6. Your module: dosing

The bottlenecks you attack

- **Blinding is rarely measured** (29.5% of 112 trials) and **often fails** (frequently above 90% in psilocybin, LSD and ayahuasca studies). When it is measured, it is usually measured once, at the end. Your agents measure it for every participant and rater, every visit, and alert within days.
- **Session burden:** two monitors per session plus a physician within 15 minutes. Your debrief agent turns spoken or typed notes into structured adverse events and unblinding cues.
- **Baseline to beat:** end-of-trial blinding check only. In the simulation, Site B raters start drifting on day 28. Report how many days your monitor takes to alert, versus 'discovered at close-out' (day 90). Also report false alarms on runs with no planted drift.

Your agents

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Session agent	Executes each scheduled dosing session; records staff present; notes protocol deviations	rec['sessions']	None	Deterministic

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Check-in agent	Runs post-session and visit check-ins (guess, confidence 0 to 100, expectancy 0 to 10) by text or voice; re-asks when an answer is ambiguous; records the channel	rec['checkins']	None	Structured form; voice via Gemini transcription if a key is set
Debrief agent	Turns monitor notes into adverse events (with severity) and unblinding cues, such as 'rater came into the room'. Escalates serious events.	rec['debriefs'], alerts	Serious adverse events go to a clinician	Regex patterns
Rater visit agent	At weeks 1, 3 and 6 records MADRS and the rater's guess, by rater ID	rec['outcomes'], rater check-ins	None	Deterministic

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Blinding monitor	Daily: asks the vault for aggregate accuracy, Bang index and u, overall, per site and per rater, for participants and raters. Decides whether a change is real drift or noise.	rec['blinding_metrics']	None	Pure statistics
Drift and action agent	When drift is detected, explains it with evidence (accuracy, interval, n, onset) and the likely cause from debriefs, and proposes a specific action (move Site B raters to remote assessment; ban session attendance)	rec['alerts'], approval request, message 'rater_drift' to master	Every action needs approval. On approval call world.apply_action so the simulation reflects it	Rules

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Missing-data chaser	Spots missed c heck-ins; reminds the parti cipant the next day; escalates to the site after two misses; proposes a site fix if a pattern appears	rec['missing_data']	Site-level fixes need approval	Rules
Literature agent	Weekly: runs the existing UNBLIND research pipeline on a new trial (or a cached c andidate when offline) and proposes adding it to the ref erence data	rec['literature'], message 'literature_update'	A human approves every addition	Cached candidates from data/gold_candidates.csv
Primary lock	Decides when every enrolled participa nt has a week-6 outcome (or the window has passed) and tells the master	message 'primary_locked'	None	Deterministic

The drift detector (your science centrepiece)

- Per rater and per site, track correct guesses over a rolling window. Compute accuracy with a Wilson 95% interval. Expected rater accuracy under working blinding is around 0.5 to 0.6.
- Fire when the lower bound exceeds 0.75 with at least 8 guesses, or use a one-sided CUSUM on correct-guess indicators against 0.55. Pick one, justify it, and report detection delay and false-alarm rate across 20 seeds.
- Participants are expected to be unblinded in psychedelic trials. Treat participant accuracy as an info-level signal that feeds the report, not as an alarm.
- Never touch allocation: the vault returns aggregates only and suppresses groups under 5. Your detector must work with that constraint. This is also your security story.

7. Your files and interfaces

You own	You read	You send	You receive
unblind/platform/dosing/* (including eval_drift.py), tests/platform/test_dosing.py	rec['enrolled'], rec['schedule'] entries with owner 'dosing', rec['measurement_plan']	rater_drift, participant_unblinding, missing_data, adverse_event, literature_update, primary_locked	protocol_amended (for example raters now remote), enrolment_complete

Danny's schedule is your input. Agree the exact schedule fields at 20:00: id, participant, site, day, kind, owner, instrument, status, and rater_id on rater visits.

8. Your task list

#	Time	Task	Priority
1	19:45	Read the contract; agree schedule and check-in shapes with Danny	Must
2	20:00	Session, check-in and rater visit agents executing the schedule (use a fake schedule until Danny's lands)	Must
3	20:30	Blinding monitor via vault.blinding_stats, stored daily	Must
4	20:50	Drift detector, alert, approval request, world.apply_action on approval, 'rater_drift' to master	Must
5	21:15	Debrief agent (regex first) and missing-data chaser	Must
6	21:30	Checkpoint: full Master run with your module	Must
7	21:45	Dosing page: live per-site and per-rater blinding chart with the drift onset marked, alert cards with approve buttons, check-in feed, debrief feed, missing-data table, KPIs	Must
8	22:30	'Play a patient' panel: a judge submits a check-in (typed; voice if time)	Should
9	22:45	Drift evaluation over 20 seeds: detection delay and false alarms vs end-of-trial check	Must
10	23:00	Integration fixes with Danny	Must
11	later	Literature agent wrapping the existing pipeline; Gemini debrief structuring	Could

9. Research tasks (when you are waiting on integration)

- Find a sourced figure for recruitment delay and screen-failure rates in clinical trials (and psychedelic trials if possible). Add it to data/bottlenecks.json with the URL and the exact quote. If you cannot source it, the slide says 'commonly reported' with no number.
- Check two rows of data/gold_candidates.csv against their sources and mark them verified with your name.

10. Tests, evaluation and judge questions

- **Tests:** metrics never come from allocation directly (grep for _arm outside world.py); groups under 5 are suppressed; drift fires after day 28 at Site B in the default world and not when drift is disabled; approval required before action; after approval drift stops; primary lock posts once.
- **Headline numbers:** median detection delay in days; false alarms per trial; check-in completion rate; debriefs structured; adverse events captured.
- **'How do you know it is drift and not chance?'** Interval-based or CUSUM rule with a stated false-alarm rate measured across seeds.
- **'Can the AI work out who got the drug?'** No. It only gets aggregates, small groups are suppressed, and every vault query is logged.
- **'Is this real data?'** No, synthetic, with planted events so we can measure the agents against a known truth.

11. Tomorrow

- Lead the judge Q&A sheet.
- Own the evaluation slide (drift detection vs end-of-trial checking).
- Present the dosing part of the demo.

12. How we work in parallel tonight

- **Contract freeze at 20:00.** After that, nobody changes the shared shapes in MODULE_CONTRACT.md without telling the other two in the group chat.
- **Own your folder.** Danny: unblind/platform/predose/, master/, core.py, world.py and app/platform_app.py. Callum: unblind/platform/dosing/. Izzy: unblind/platform/postdose/ and eval/platform/. Nobody edits another person's folder.
- **Git.** One branch each (danny/predose, callum/dosing, izzy/postdose). Pull main at every checkpoint. Danny merges.
- **Every module works with no Gemini key.** Deterministic fallback first, Gemini second. The demo must survive bad Wi-Fi.
- **Every agent explains itself.** rec.event(agent, text, level, module) for each decision. Decisions are what make it look and behave like an agent.
- **Human gate.** Anything that changes the trial goes through the ApprovalQueue.
- **Vault only.** Never compute anything from allocation outside the vault.
- **Label synthetic data** on every page and in every chart.
- **Copy:** UK English, plain language, no em dashes.

Tonight's timeline

Time	Everyone	Checkpoint
19:45 to 20:00	Read this plan. Agree the contract. Danny sets up branches and the scaffold.	Contract frozen
20:00 to 21:30	Build block 1: agents with deterministic fallbacks, plus offline tests	Each module runs inside the master with the others present or missing
21:30 to 21:45	Checkpoint call: run Master end to end on main	Known bugs listed
21:45 to 23:00	Build block 2: module page, KPIs, Gemini enhancements, your demo moment	Pages render in mission control
23:00 to 23:30	Integration: Danny runs the full demo path twice; others fix their bugs	Clean run from setup to close-out
23:30 to 00:00	Freeze. One rehearsal of the demo path. Push and tag v1.	Working prototype on main

Tomorrow

Task	Lead	Support
One-page write-up (problem, who it helps, data, evaluation, limits and safeguards)	Izzy	Callum (evaluation numbers), Danny (architecture)
Pitch deck and 4-minute script	Danny	Izzy (science slides), Callum (evaluation slide)
Judge Q&A sheet (sense checks above, with answers)	Callum	Izzy
Demo rehearsal x3, including a judge's unexpected input	Danny	All
README and repo tidy	Danny	All

13. Demo script (4 minutes + 2 minutes for judges)

Time	Speaker	On screen
0:00 to 0:30	Izzy	The problem: patients know they tripped. 29.5% of trials measure blinding; many fail above 90%. FDA July 2026 names it the central threat.
0:30 to 1:30	Danny	Pre-dosing: paste a draft protocol. Risk score, ranked fixes, FDA checklist, drafted blinding section. Approve a fix and watch it re-check. 260 applicants screened to a shortlist; flagged free-text disclosures shown with quotes.
1:30 to 2:40	Callum	Press play. Live blinding chart per site. Site B's raters drift; the agent explains and proposes remote raters; approve; drift stops. Show a debrief turned into structured events.
2:40 to 3:30	Izzy	Post-dosing: researcher asks a leading question; the bias gate rewrites or blocks it. Close-out FDA blinding report: reported vs expectancy-adjusted effect, with every decision traceable.
3:30 to 4:00	Danny	What it means: one standard, every trial comparable, a dataset that improves every new trial. Next step: pilot with one CRO.
4:00 to 6:00	Judges	They edit the protocol, play a patient, or submit a question. Everything is live.

14. Glossary

Term	Plain meaning
Blinding	Keeping secret who got the real drug
Functional unblinding	The drug's effects give the secret away
Expectancy	Feeling or scoring better because you believe you got the drug
Guess accuracy, u	Share of correct 'which group?' guesses. u = 2 x accuracy - 1, so 0 is a coin flip and 1 means everyone knew
Bang blinding index	Standard blinding score per arm, from -1 to 1, where 0 is ideal
Central blinded rater	The person scoring the patient is remote, never sees the session, and does not know the arm
Stratified randomisation	Randomising separately within groups (for example, experienced vs naive), so both arms get the same mix
MADRS	A clinician-scored depression scale, 0 to 60, higher is worse
Primary endpoint and lock	The one outcome the trial promised to test; locking it means no more changes before analysis
Hedges' g	Effect size in standard-deviation units, comparable across scales
Meta-regression	A line fitted across many trials, one dot per trial
CRO	Contract research organisation: companies that run trials for drug makers
Synthetic data	Simulated data, labelled as such, never presented as real-world performance